A Palaeomagnetic Study of some Lower Jurassic Rocks of N.W. Europe

R. W. Girdler

(Received 1959 January 8)

Summary

A study has been made of the directions of natural remanent magnetization of the Upper Lias Sands of England (Section 1) and reasons are given for considering the magnetization of some of these to be stable. The results show that the direction of the Earth's magnetic field in Lower Jurassic times was significantly different from that of the present Earth's field, having an easterly declination and inclination 64° to 70°.

The work was extended (Section 2) to include some Lower Jurassic volcanic rocks from Southern France and these gave results in general agreement with those for England.

It is concluded (Section 3) that the ancient geomagnetic pole position for the Lower Jurassic period lies in the region of the Caspian Basin.

1. A palaeomagnetic study of some Lower Jurassic sedimentary rocks of Great Britain

So far, few palaeomagnetic results have been obtained for rocks of Jurassic age from Great Britain. Nairn (1956, 1957) has given results for Jurassic material collected from North Yorkshire and Scotland. The estuarine sediments of Yorkshire have directions of magnetization close to the present Earth's field and Nairn (1957) gives reasons for considering these to be unstable. From Scotland, three limestone samples from the base of the Estuarine Beds of Skye and one sandstone sample from the Brora Arenaceous Series were found to be reversely magnetized.

The Upper Lias Sands are known as the Bridport, Yeovil, Midford and Cotswold Sands and the Blea Wyke Beds. The first four form one unit in south-west England but are only exposed at the localities from which they are named. The Blea Wyke Beds are exposed on the Yorkshire coast.

The age relationships were determined by S. S. Buckman (1889, 1910) and are summarized in Figure 1. The mineralogy of the sands has been described by P. G. H. Boswell (1924).

It was concluded that the sands are suitable for palaeomagnetic work for the following reasons:

(a) They occur over a wide area and are well dated.

(b) The uniformity and fineness of grade suggest that they were deposited in quiet water, free from violent current and wave action.

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022

(c) The sands have remained fresh and unaffected by subsequent geological events and it is possible that they have an undisturbed "depositional" remanent magnetization.

The stratigraphical horizons sampled are shown approximately in Figure 1 by the letters *N* or *R*, the letters referring to horizons found to have normal or reverse directions of magnetization respectively.

Two or three specimens were cut from each sample. The specimens were in the form of cylinders with height equal to diameter (equal to 3·5 cm) and a special perspex holder was constructed so that a cylinder could be measured in six positions on-centre beneath the astatic magnetometer. The six positions correspond to placing the six faces of a cube in turn beneath the magnet system. The direction

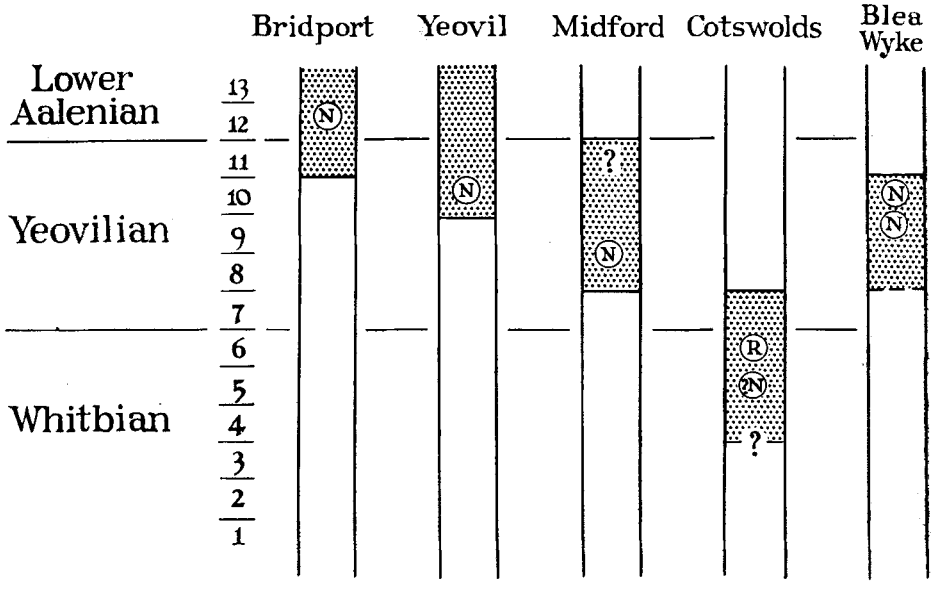


FIG. 1.—Age relationships of the Upper Lias diachronous sands (adapted from Arkell 1933). The numbers refer to the ammonite subzones given by Dr W. J. Arkell.

of magnetization was computed from the sums ΣX, ΣY, ΣZ of the eight deflexions for the x, y, z directions of the orthogonal co-ordinate reference system chosen for the cylinder.

The directions of natural remanent magnetization for all the specimens measured are plotted on Schmidt equal area projections. Figures 2 and 3 show the directions of magnetization of the Bridport and Yeovil Sands respectively. The directions for the Midford Sands are shown in Figures 4a and 4b; for the Midford Sands it was necessary to correct for the tilt of the strata and Figures 4a and 4b show the results before and after applying the geological corrections. Figures 5a and 5b show the directions for the Cotswold Sands; some of these are reversely magnetized (Figure 5a) and some are normally magnetized (Figure 5b). The Blea Wyke Beds of Yorkshire consist of grey, micaceous sandstone (38 ft) overlain by soft, yellow sandstone (30 ft); the directions of magnetization of these are shown separately in Figures 6a and 6b respectively. The mean directions for all the results are illustrated in Figure 7.

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022

It is evident from Figure 7 that the Upper Lias Sands show varying degrees of stability of magnetization. The Yeovil Sands and the Blea Wyke grey sands are magnetized in directions close to the Earth's present field and are presumed to be unstable; the normally magnetized Cotswold Sands, the Blea Wyke yellow sands and the Bridport Sands have directions of magnetization ranging from 6° to 23° to the east of the Earth's present field and it is suggested that these are partially stable; the reversely magnetized Cotswold Sands and the normally magnetized Midford Sands have a natural remanent magnetization in an east–westerly direction and it is likely that these are stable.

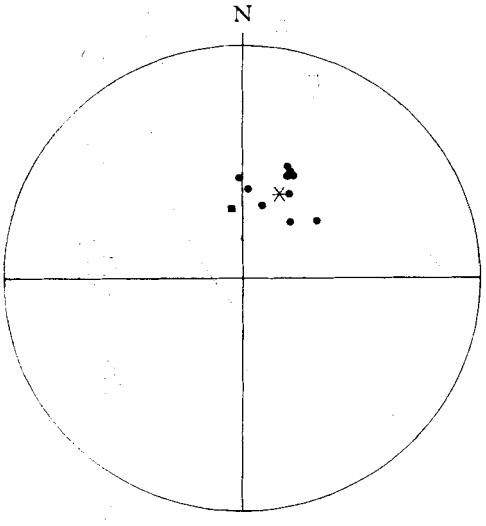


FIG. 2.—Directions of natural remanent magnetization of the Bridport Sands.

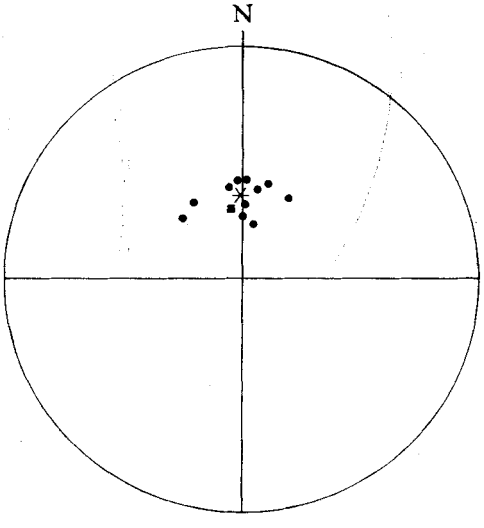


FIG. 3.—Directions of natural remanent magnetization for the Yeovil Sands.

In this and succeeding diagrams the plane of the projection (Schmidt equal area) is the equatorial plane; normal directions of magnetization (downwards) are shown by dots and reverse directions of magnetization (upwards) are shown by circles; the mean direction is shown by an asterisk and the direction of the Earth's present magnetic field by a square.

The following reasons are given for considering the reversely magnetized Cotswold Sands and the normally magnetized Midford Sands to be stable:

(a) The occurrence of a reverse direction of magnetization is in itself evidence of stability.

(b) Figures 4a and 4b show that the directions of magnetization for the Midford Sands become less scattered after geological corrections have been applied (Graham's stability test, 1949). This suggests that the magnetization was acquired before tilting and that the direction of polarization has since been preserved.

(c) The mean direction of magnetization for the normally magnetized Midford Sands is slightly further from the Earth's present field than the axis of magnetization of the reversely magnetized Cotswold Sands suggesting that the Midford Sands are at least as stable as the reversely magnetized Cotswold Sands.

(d) Several specimens were re-measured after a period of six months after having been stored in random directions in the Earth's field No significant changes in the directions of natural remanent magnetization were observed.

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022

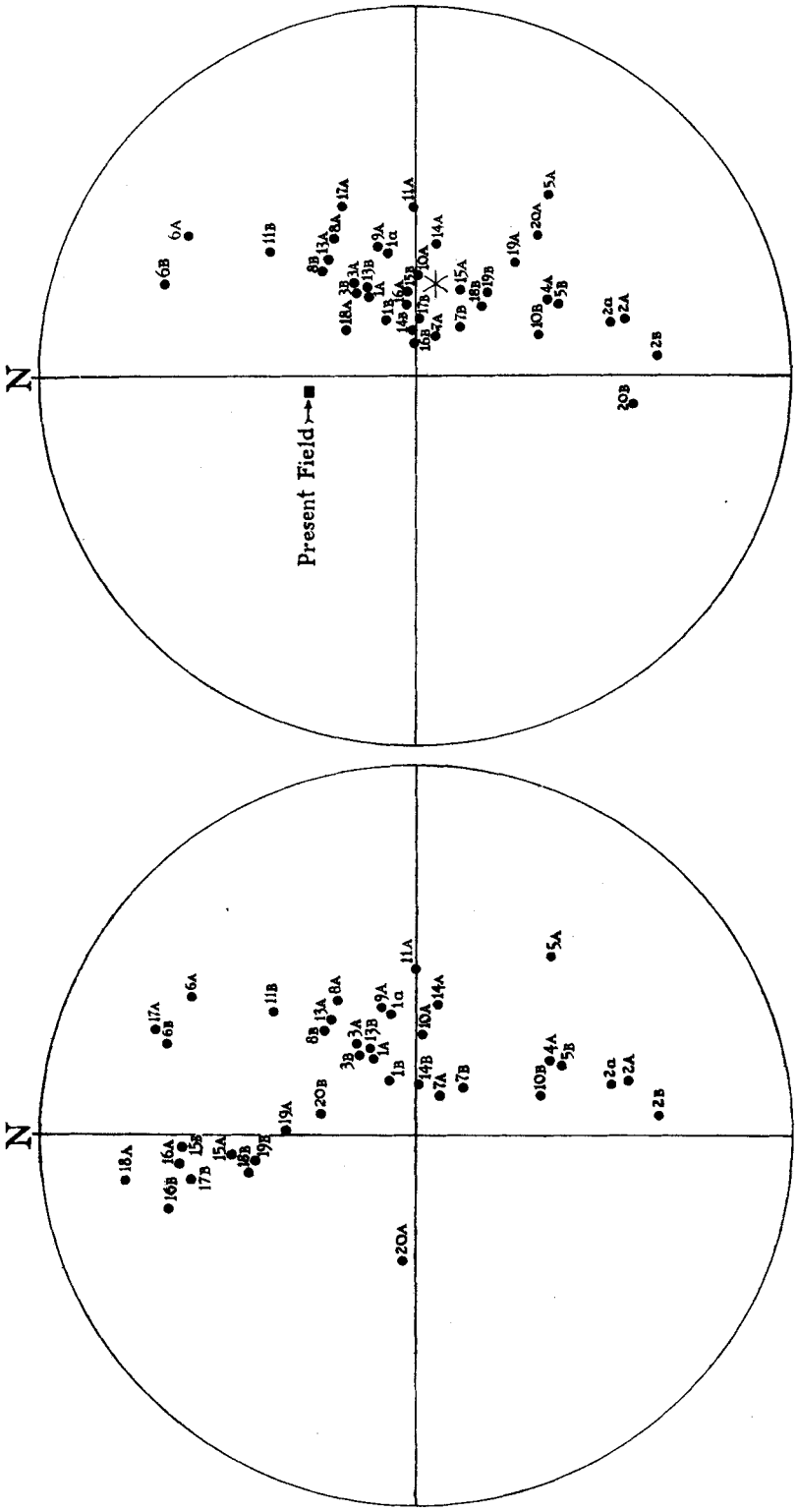


FIG. 4a.—Directions of natural remanent magnetization for the Midford Sands.

FIG. 4b.—Directions of natural remanent magnetization for the Midford Sands after correction for geological dip.

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022

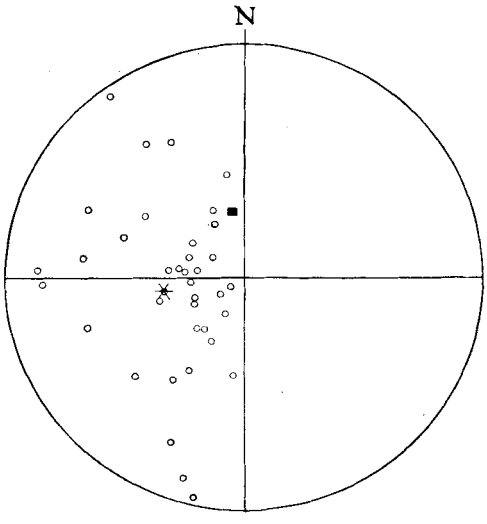


FIG. 5a.—Directions of natural remanent magnetization for the Cotswold Sands—reversely magnetized specimens.

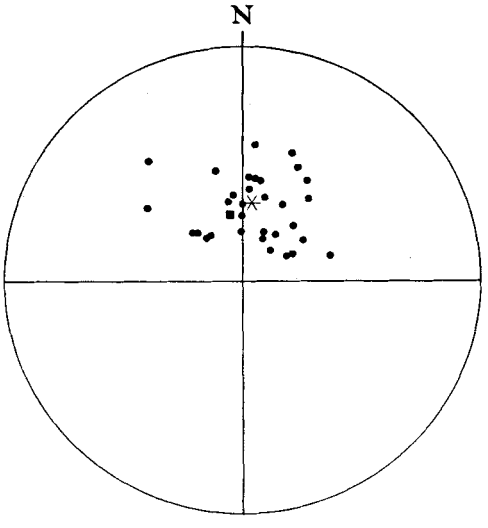


FIG. 5b.—Directions of natural remanent magnetization for the Cotswold Sands—normally magnetized specimens.

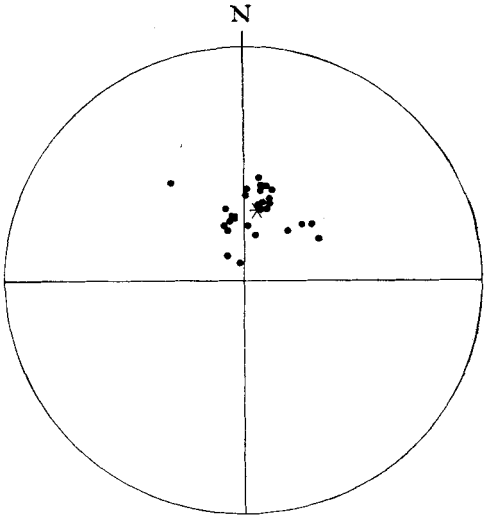


FIG. 6a.—Directions of natural remanent magnetization for the Blea Wyke Beds—yellow sands.

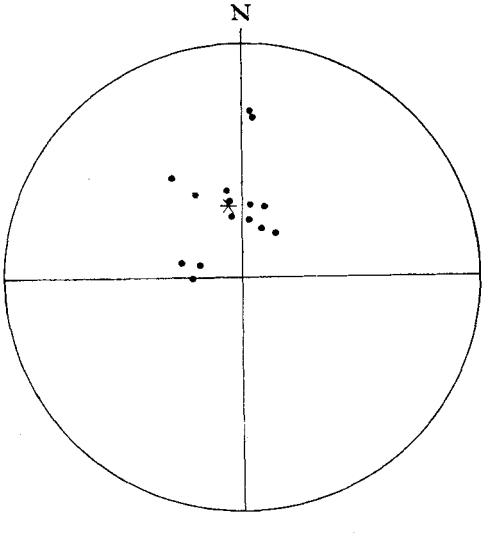


FIG. 6b.—Directions of natural remanent magnetization for the Blea Wyke Beds—grey sands.

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022

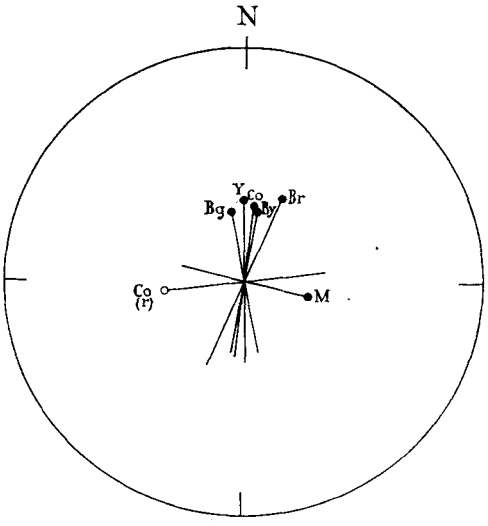


FIG. 7.—Mean directions of natural remanent magnetization for the Upper Lias Sands.

Br = Bridport Sands.
Y = Yeovil Sands.
M = Midford Sands.
Co = Cotswold Sands—normally magnetized samples.
Co(r) = Cotswold Sands—reversely magnetized samples.
By = Blea Wyke Beds—yellow sands.
Bg = Blea Wyke Beds—grey sands.

The values for the mean directions of natural remanent magnetization and the Fisher statistics (Fisher 1953) are given in Table 1.

Table 1

Mean directions of magnetization for the Upper Lias sands of England

		θ °	ϕ °	α °	K	N	S
Bridport Sands		23·4	60·4	5·1	73·1	12	7
Yeovil Sands		358·9	63·9	6·1	57·3	11	10
Midford Sands		103·5	69·8	7·4	10·2	42	20
Cotswold Sands	N	6·5	65·5	5·5	22·1	32	17
	R	262·6	−64·1	9·9	6·5	38	17
Blea Wyke Beds	Yellow	9·7	67·0	4·2	46·0	26	9
	Grey	349·2	66·3	9·1	18·4	15	5
Hettangian tuffs		54·9	58·9	5·8	17·0	38	18

θ = Declination, measured clockwise from north.
ϕ = Inclination, with reference to the horizontal plane.
(positive inclination—downwards negative inclination—upwards.)
α = Radius of 95 per cent circle of confidence (Fisher 1953).
K = Fisher's precision factor.
N = Number of cylinders measured.
S = Number of rock samples.

Figures 4b and 5a show that the normally magnetized Midford Sands and the reversely magnetized Cotswold Sands have a larger scatter in the directions of magnetization For this reason, the sampling was extended and the mean directions of magnetization have been calculated from 42 specimens (20 samples) for

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022

the Midford Sands and 38 specimens (17 samples) for the reversely magnetized Cotswold Sands. The following are possible reasons for the large scatter:

(a) It is possible that some of the sands have been disturbed due to movement of the underlying Upper Lias Clays.

(b) In some cases, there is a possibility that the sands have been disturbed by vegetation, especially where collection was made from sunken lanes.

(c) The fine grade of the sands and the absence of any obvious bedding planes makes it difficult to ascertain accurate geological corrections.

The results show that some Lower Jurassic sediments of Great Britain have a mean direction of polarization with declination N. 83–103° and inclination 64–70°.

Table 2 shows a comparison with Nairn's (1957) results.

Table 2

Comparison with Nairn's results

	θ °	ϕ °	α °	K	N	S
Nairn (1957)	234·4	−66	6·6	33·5	14	4
Midford Sands	283·5	−69·8	7·4	10·2	42	20
Cotswold Sands	262·6	−64·1	9·9	6·5	38	17

(The symbols have the same meaning as in Table 1)

The main differences are in the values for the declination which are 28° to 49° larger for the Cotswold and Midford Sands than for Nairn's rocks.

Nairn's specimens are mainly from the Middle Jurassic and are therefore slightly younger in age than the Upper Lias Sands. However, the comparison may not be justified as Nairn's results depend on measurements from only four samples and no stability tests were applied.

2. Directions of magnetization of some Lower Jurassic Volcanic rocks from France

A collection has been made of some basalts and tuffs of Lower Hettangian age from the northern foothills of the Pyrenees, the object being to compare the directions of natural remanent magnetization with those obtained for the Lower Jurassic sedimentary rocks of England.

Results are given for volcanic rocks collected from an area west of Foix (Latitude 42°57′ N, Longitude 1°35′ E). In this area volcanic tuffs occur in the middle of the Hettangian Limestone and are therefore well dated. The tuffs contain fragments of basalts, limestones and marls and are of variable texture. The thickness varies from 1·5 to 10 m. Stratigraphical details have been given by Dubar (1925).

The volcanic rocks of the northern Pyrenees are equivalent in age to the Lower Lias of Great Britain. The stratigraphical relations with respect to the Upper Lias Sands discussed in Section 1 are shown in Table 3.

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022

Table 3

Age relations of Upper Lias sands and Lower Lias volcanic rocks of Pyrenees

Stage	Formation	Facies
Toarcian	Upper Lias	Bridport Sands Yeovil Sands Midford Sands Cotswold Sands
Upper Pliensbachian	Middle Lias	
Lower Pliensbachian Sinemurian Hettangian	Lower Lias	Volcanic rocks of north Pyrenees

As the whole of the Jurassic material was laid down in approximately 40 million years, from 140 to 180 million years ago (Mayne & others 1959) the maximum age difference between the Hettangian tuffs and the Upper Lias Sands is likely to be less than 12 million years.

Eighteen orientated samples were collected from five sites west of Foix. Unfortunately, no time was available for geological mapping which was necessary to ascertain accurate corrections for the dip of the strata. The geological corrections have been estimated with the aid of field notes and survey maps and these have been applied to the directions of natural remanent magnetization before computation of the mean directions of magnetization given in Table 1. There is likely to be little error in the values used for geological strike but it is possible that some of the corrections for geological dip may be in error by $\pm 15°$.

Specimens were cut in the form of cylinders and measurements made as described in Section 1.

The mean directions of magnetization together with the Fisher statistics are given in Table 4. The directions for all the samples are illustrated in Figures 8a and 8b: Figure 8a shows the computed directions and Figure 8b the directions after the geological corrections have been applied.

Table 4

Mean directions of magnetization for the Lower Lias volcanic rocks of France

	θ °	ϕ °	α °	K	N	S
Hettangian tuffs	54·9	58·9	5·8	17·0	38	18

The rocks are considered to have a stable magnetization for the following reasons:

(a) Nearly all the specimens are magnetized in a south-easterly direction (Figure 8a), far from the direction of the Earth's present field.

(b) When geological corrections are applied the directions of magnetization become less scattered (Figure 8b) and this constitutes an application of the stability test of J. W. Graham (1949).

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022

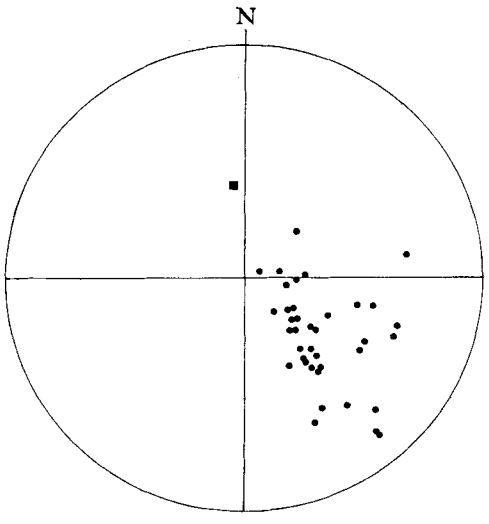


FIG. 8a.—Directions of natural remanent magnetization for Hettangian tuffs from the northern Pyrenees Central area.

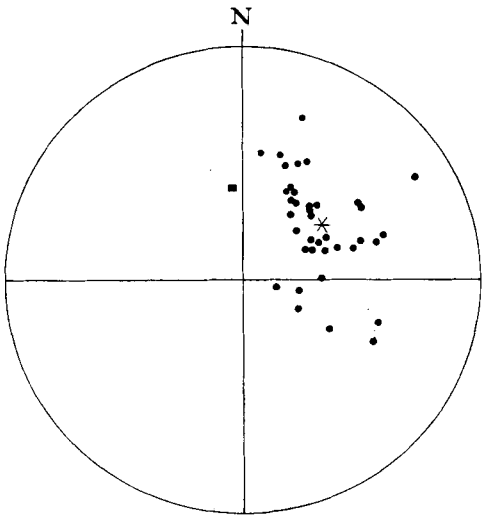


FIG. 8b.—Directions of natural remanent magnetization for Hettangian tuffs from the northern Pyrenees Central area after geological corrections have been applied.

3. Geomagnetic pole positions for the Lower Jurassic period

Pole positions have been computed from the directions of natural remanent magnetization of the Midford Sands, the reversely magnetized Cotswold Sands and the Hettangian volcanic rocks. The lengths of the semi-major and semi-minor axes of the confidence ovals have been calculated from the Fisher radius of confidence at the 5 per cent level by the method suggested by Creer & others (1957). These are given in Table 5 and illustrated on a stereographic projection for the northern hemisphere in Figure 9.

Table 5

Pole positions for the Lower Jurassic

	Pole position		Confidence Oval	
	Lat.	Long.		
Midford Sands	32°·8N	41°·0E	12°·8	11°·0
Cotswold Sands	38·2	59·3	15·7	12·5
Pyrenees Central area	49·0	76·6	8·6	6·4

It is seen from Figure 9 that the pole positions computed from two completely different rock types from different areas show reasonably good agreement and it is concluded that the geomagnetic pole for the Lower Jurassic period as calculated from rocks from Great Britain and France lies in the general region of the Caspian Basin.

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022

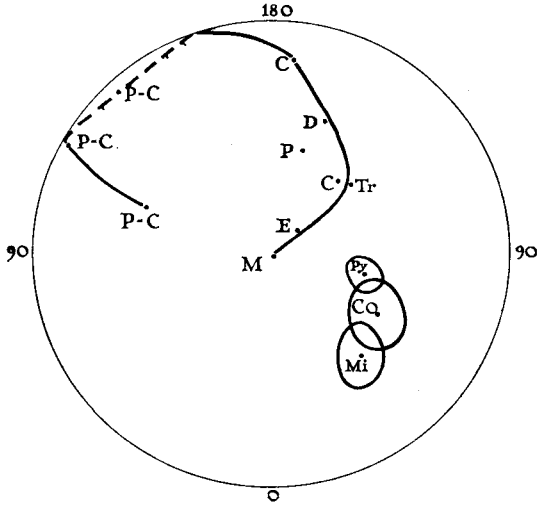


FIG. 9.—Ancient geomagnetic pole positions for the Lower Jurassic period.

Mi = Pole position computed from mean direction of magnetization of the Midford Sands.
Co = Pole position computed from mean direction of magnetization of reversely magnetized Cotswold Sands.
Py = Pole position computed from mean direction of magnetization of Hettangian tuffs from the Pyrenees Central area.

The polar wander curve given by Creer & others (1957) is also shown. The symbols in stratigraphical order are: M = Mioccene, E = Eocene, Tr = Trias, P = Permian, C = Carboniferous, D = Devonian, C = Cambrian, P-C = Pre-Cambrian.

4. Acknowledgments

The work reported on above was carried out at the Department of Geodesy and Geophysics of the University of Cambridge.

My sincere thanks are due to:

Mr B. C. Browne, Mr J. C. Belshé, Dr A. E. M. Nairn (King's College, University of Durham) and Mr R. Wilson (Imperial College, London) for helpful discussions.

Dr D. V. Agar (Royal School of Mines, London), Mr E. Cameron (University of Manchester), Dr D. T. Donovan (University of Bristol) and Dr G. A. Kellaway (H.M. Geological Survey) for map references where suitable exposures of the Upper Lias Sands were found.

Mr J. C. Belshé, Mr D. R. Reeves, Mr G. A. Day and Mr M. J. Keen for their help on rock collecting expeditions.

The Royal Dutch Shell Oil Company for a Research Studentship which I was privileged to hold for three years.

References

Arkell, W. J., 1933. *The Jurassic System in Great Britain* (Oxford University Press).

Boswell, P. G. H., 1924. *Geol. Mag., Lond.*, **41**, 246.

Buckman, S. S., 1889. *Quart. J. Geol. Soc. Lond.*, **45**, 440.

Buckman, S. S., 1910. *Quart. J. Geol. Soc. Lond.*, **66**, 80.

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022

Creer, K. M., Irving, E. & Runcorn, S. K., 1957. *Phil. Trans. Roy. Soc. A*, **250**, 144.
Dubar, G., 1925. *Mém. de la Soc. Géol. du Nord.*, **9**, 1.
Fisher, R. A., 1953. *Proc. Roy. Soc. A*, **217**, 295.
Graham, J. W., 1949. *J. Geophys. Res.*, **54**, 131.
Mayne, K. I., Lambert, R. St. J. & York, D., 1959. *Nature, London*, **183**, 212.
Nairn, A. E. M., 1956., *Nature, London*, **178**, 935.
Nairn, A. E. M., 1957. *Phil. Mag. Suppl. Adv. in Phys.*, **6**, 162.

Downloaded from https://academic.oup.com/gji/article/2/4/353/662671 by Curtin University Library user on 15 September 2022